

CS101

Command Line

1/15

Agenda

1. Teacher/Student Contract
2. Syllabus
3. Lecture: Into to Command Line Interface (CLI)
4. Group Demo: CLI
5. Work on Lab1
6. Touch Typing Initial Test

10 min break at ~5pm

Student-Teacher Contract

Syllabus

(will be on canvas)

Syllabus Highlights

- No penalty up to 3 missed class days (starting next week)
- Labs are pass/fail, no late submissions
- 3 free late days for HWs (total)
- 13 Quizzes, drop worst 3. No makeups

Lecture: Intro to CLI

Computer Basic Diagram

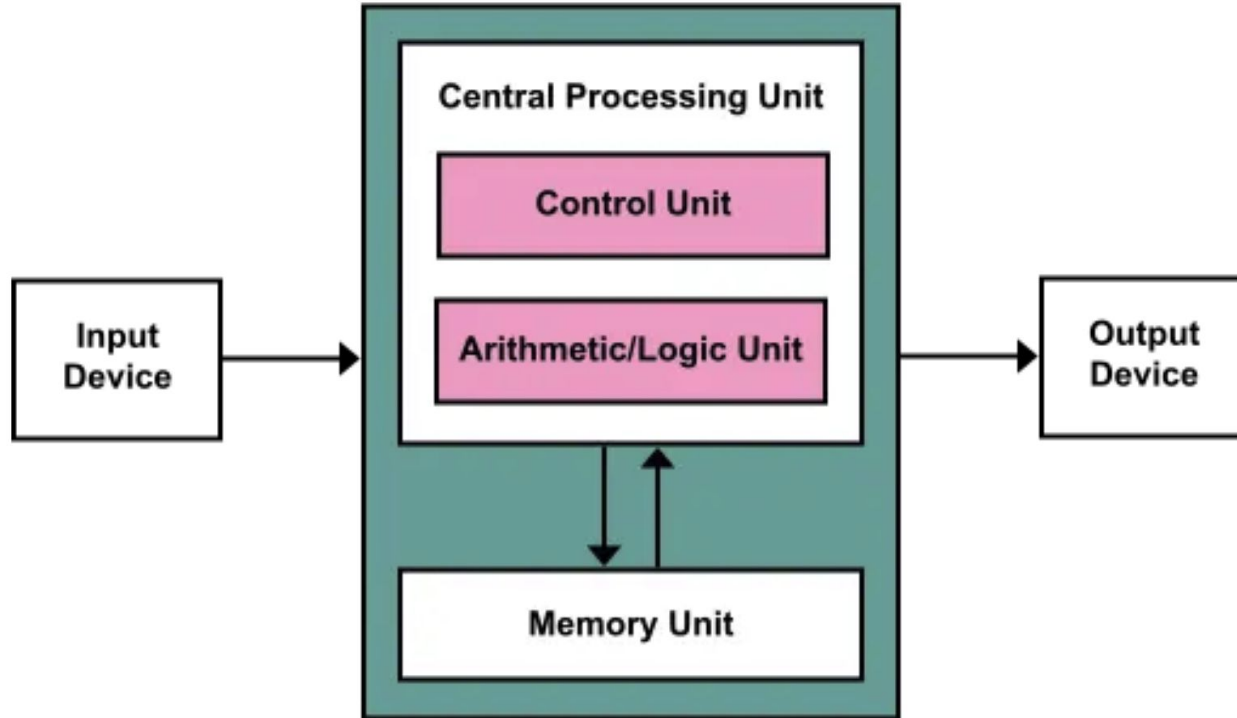
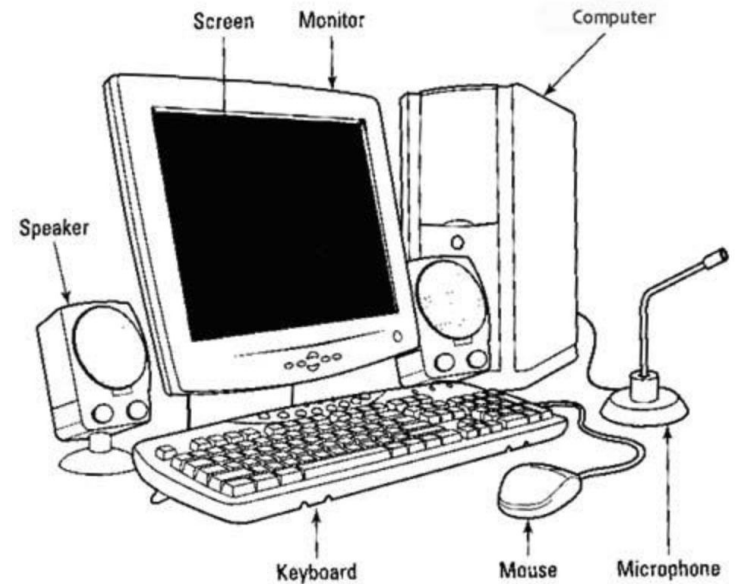


Image: <https://medium.com/understanding-computer-science/the-anatomy-of-a-computer-part-1-of-4-4fda1928c774>

Inputs → CPU → Outputs



First Personal Computer (PC)

- 1981 – IBM 5150
- \$5,000 to \$15,000 (adjusted)
- No mouse! Keyboard only



Apple Macintosh 1984

- First “affordable” PC with a mouse
- \$7,000 (adjusted)



Group Demo: CLI

Why we are learning CLI

- Makes debugging computer issues easier
- Some software only available via CLI
- Remote machines typically only have CLI

Before we Start

- Terminal → Application with Command Line Interface
 - Some Terminals:
 - Powershell → windows
 - Bash → unix systems (chromebooks, linux, macOS)
- To “run” a command means “press enter”
- In CLI, everything is **CASE SENSITIVE**
- CLI requires **precision** (no typos)
- Show in File Explorer:
 - **folder** aka **directory** aka **dir** (same thing)
 - **subdir** aka **child**
- (Pass out Cheatsheet and slide sheet)

Interact with computer using Bash

1. Open File Explorer and select
 - a. Linux → Ubuntu → home → computerXXX
 - b. See file updates visually as you run commands in bash
2. Open “Ubuntu” App (bash)

Run (one at a time):

1. `pwd`
2. `mkdir subdir1`
3. `cd subdir1`
4. `touch file1.txt`
5. `touch .secret1.txt`
6. `ls`
7. `ls -a`
8. `ls -a -l`
9. `ls -al`
10. `ls -la`
11. `ls --help`
12. `rm .secret1.txt`
13. `cd ..`
14. `rm -r subdir1`

Tips

- Syntax for commands (note spaces between):
 - `cmd -options input1 input2 input3 inputX`
- To see command options etc, use
 - `cmd --help`
 - For example
 - `ls --help`
- Experiment with pressing “tab”! Saves a lot of typing
- Forward slash, “/”, separates parent/child folder
- Period, “.”, (usually) means “in current directory”
 - Period in front of file means “secret” file, ie `“.bash_history”`

Codespaces

Remote Machine with Bash terminal

In TERMINAL, run command:

- `curl ipinfo.io`

Lab1 Setup

[https://github.com/icanhazcodeplz/CS101/tree/main/labs/lab1
_shell_script](https://github.com/icanhazcodeplz/CS101/tree/main/labs/lab1_shell_script)

Next Week

- Zoom Office Hours Tomorrow
- Before next class (Homework)
 - Continue tutorials and/or video for [Topic 1: Command Line \(CLI\)](#)
 - Submit first attempt of Lab1 on Canvas if possible
 - Does not need to be perfect! (unlimited submissions)
 - Email me if stuck
- During Next Class
 - Lecture: Binary; compilers
 - Java HelloWorld Demo
 - Start Lab2
- Teaser (soon)
 - Take apart a laptop!